

Ben-Gurion University of the Negev

Faculty of Engineering Science

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Fourth-Year Engineering Project

Preliminary Report

MagViVit: Transformer-based spatiotemporal magnification for vital signs monitoring

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1.a. תקציר

ניטור רציף ביחידות לטיפול נמרץ פגים (NICU) הוא חיוני, אך הפרקטיקה הסטנדרטית מסתמכת על חיישנים פיזיים. חיישנים אלו גורמים לנזקי עור כואבים (MARSI), לסטרס נוירו-התפתחותי הפוגע בהתפתחות המוח, וסובלים מאמינות נמוכה עקב רגישות לארטיפקטים תנועתיים. החידוש המוצע הוא פיתוח מודל למידה עמוקה רב-ערוצי (Multi-modal) ללא מגע, המבוסס על עיבוד מקבילי (Parallel Processing) של וידאו IR ו-NIR, לחיזוי רציף ומדויק של טמפרטורת גוף וקצב נשימה. החידוש טמון בארכיטקטורה היברידית דו-ענפית (Dual-Branch CNN-Transformer). מודל זה משתמש ב-NIR לזיהוי אנטומי וניתוח תנועה, וב-IR ו NIR למדידת טמפרטורה באופן עקיף וזיהוי זרימת אוויר, ליצירת מערכת יתירה (Redundant) ואמינה. השיטה כוללת רישום תמונות (IR-NIR), חילוץ תכונות מרחבי (CNN), ומידול זמני (Transformer) לניתוח תלויות ארוכות טווח וזיהוי אנומליות פיזיולוגיות. תוצאות צפויות: מודל Proof-of-Concept המדגים היתכנות לניטור בדיוק קליני, המאפשר זיהוי מוקדם של התדרדרות ומפחית נזק יאטרוגני.

מילות מפתח: טיפול נמרץ פגים (NICU), ניטור ללא מגע, למידה עמוקה, היתוך רב-ערוצי (Multi-modal Fusion), וידאו NIR, וידאו IR, חיזוי מדדים חיוניים, מידול זמני (Transformer), קצב נשימה, טמפרטורת גוף.

1.a. Abstract

Continuous monitoring in the Neonatal Intensive Care Unit (NICU) is vital for preterm infants, yet standard practice relies on adhesive sensors. These sensors cause painful skin injuries (MARSI), neurodevelopmental stress linked to brain dysmaturation, and suffer from low reliability due to motion artifacts. The new method is to develop a contactless, multi-modal deep learning model, based on parallel processing of IR and NIR video, for the continuous and accurate prediction of body temperature and respiratory rate. The innovation lies in a hybrid, dual-branch CNN-Transformer architecture. This model leverages NIR for anatomical context and motion analysis, while utilizing NIR and IR for non-direct thermography and airflow detection, creating a robust and redundant system. The method involves image registration (IR-NIR), spatial feature extraction (CNN), and temporal modeling (Transformer) to analyze long-range dependencies and identify physiological anomalies. Expected Results: A Proof-of-Concept (PoC) model demonstrating clinical-grade accuracy, enabling early detection of deterioration (e.g., sepsis) and reducing iatrogenic harm.

Keywords: Neonatal Intensive Care Unit (NICU), Contactless Monitoring, Deep Learning, Multi-modal Fusion, NIR Video, IR Video, Vital Sign Prediction, Respiratory Rate, Body Temperature, Transformer.

2. Research Proposal

a. Essence of the Product

The essence of this project is the development of a Proof-of-Concept (PoC) software model. This model will be based on a deep learning architecture capable of continuous, non-contact prediction of vital signs for preterm infants within the Neonatal Intensive Care Unit (NICU).
The product will be engineered to ingest and process synchronous video streams. During model training, both NIR video and its converted IR representation will be used, while the final deployed system will operate using only NIR video.
The model's primary output will be a continuous, real-time numerical prediction of two critical vital signs: **Body Temperature (T)** with **MSE of 0.2 Celsius degrees** and **Respiratory Rate (RR)** with **MSE of 1.6 BPM.**

b. Performance Specifications

The primary objective of the PoC model is to maximize accuracy in vital sign prediction. The system's output will be rigorously validated against a "Ground Truth" dataset extracted from clinical monitors. Performance will be benchmarked against current state-of-the-art (SOTA) non-contact monitoring methods, aiming for low Mean Absolute Error (MAE) and high correlation with medical-grade contact sensors.

c. The Technology

The proposed technology is a dual-branch, multi-modal architecture that combines Convolutional Neural Networks (CNNs) and Transformers in two parallel branches.

- **CNNs as Spatial Feature Extractors:** Two parallel CNN pipelines will operate on the streams. The RGB-CNN branch (for RR) will use a Pretrained Autoencoder CNN for Video Motion Magnification and a ResNet-50 to extract motion features from Grayscale Frames. The NIR-CNN branch (for Temp) will use a Pretrained UNet to generate Estimated IR maps and a Face-aware ResNet-50 to extract thermal patterns.
- **Transformers as Temporal Modelers:** Two separate Transformer Encoder blocks receive sequences of Token Embeddings (derived from their respective CNNs via an FC layer) along with Positional Embeddings. Their self-attention mechanisms are uniquely suited to model long-range temporal dependencies. This allows them to learn complex patterns across different time scales, from the rapid, short-scale rhythm of respiration (seconds) to the slow.

d. Methodology and Research Phases

The project will be executed in a structured, five-phase pipeline:

1. **Data Acquisition and Calibration:** This foundational phase includes acquiring the clinical dataset of synchronized RGB and NIR videos from the NICU. It involves two critical sub-tasks:
    - **Ground Truth Generation:** This task is crucial for generating the ground truth data and utilizes a hybrid approach: A calibrated RGB camera focused on the monitor, leveraging classical computer vision algorithms (such as edge detection, template matching, or digit classification like MNIST) for initial data extraction. This extracted data will then be synchronized with the video stream.
    - **Image Registration:** A precise calibration process to compute the transformation matrix that spatially aligns the RGB and NIR video streams, likely optimized by maximizing Mutual Information (MI) between the two modalities.
2. **Video Pre-processing:** Preparing the raw video data for the model. This includes using CNNs for object detection (to isolate torso + face regions) and semantic skin segmentation and applying a Video Motion Magnification model (Pretrained Autoencoder CNN) to the NIR stream to amplify subtle respiratory motions. Simultaneously, the NIR stream will be processed by a Pretrained UNet to generate Estimated IR maps.
3. **Data Augmentation:** Implementation of an extensive data augmentation strategy to ensure model robustness. This will include spatial augmentations (rotation, cropping), temporal augmentations, and the addition of synthetic noise and motion artifacts to simulate realistic NICU conditions.
4. **Hybrid Model Development:** This is the core research task of building and training the dual-branch architecture. Pre-trained CNNs (ResNet-50 for NIR stream for both branches) will serve as spatial feature extractors for each frame, and two separate Transformer Encoder architectures will act as temporal modelers to analyze the sequence of these features (as Token Embeddings) over time.
5. **Validation:** The trained model will be rigorously evaluated against a held-out test set to ensure it meets the predefined performance specifications.

3. Project Definition & System Architecture

To define the scope of the research and align expectations, the project is defined by the following deliverables and boundaries:

- **Deliverables:**
  - **Software Model (PoC):** A complete, executable software pipeline capable of ingesting video files and outputting estimated vital signs.
  - **Deep Learning Architecture:** A trained and validated dual-branch neural network (CNN + Transformer) for the estimation of Respiratory Rate (RR) and Body Temperature.
  - **Validation Report:** A comparative analysis of performance against Ground Truth data, including statistical error metrics (MAE, RMSE).
- **Project Boundaries (Scope):**
  - **In Scope:** Processing of pre-recorded video data (offline), single-subject tracking, and software-level implementation of deep learning modules.
  - **Out of Scope:** Hardware development (sensors/cameras), real-time embedded optimization.
- **High-Level System "Black Box":**
  - **Inputs:** Raw NIR Video Stream + Synchronized Ground Truth (for training).
  - **System:** "MagViVit" Deep Learning Architecture.
  - **Outputs:** Continuous numerical predictions for Respiratory Rate (RR) and Body Temperature(T).

Figure 1: **System's Architecture**, written by Maor Oz from our research crew, rotated 90 degrees

The system's workflow begins with a preprocessing pipeline where Torso Face Detection and a Sort Algorithm (using Data Association and KF Prediction) generate Torso + Face Frames and Face Masks Frames.

The preprocessed frames are fed into a neural network split by modality:

- **Respiratory Rate (RR)** Branch: Uses the NIR stream. It applies Video Motion Magnification (via a pretrained Autoencoder CNN) and converts the frames to Greyscale. A shared ResNet-50 extracts spatial features.
- **Temperature (Temp)** Branch: Uses the NIR stream. It applies a pretrained UNet to create estimated IR maps. These maps, along with the Face Masks Frames, are fed into a Face-aware ResNet-50 for feature extraction.

Following spatial extraction, both branches convert their features to Token Embeddings, add Positional Embeddings, and use a shared Transformer Encoder for temporal modeling. Finally, a distinct FC + LayerNorm block in each branch produces the continuous predictions for RR and Temp.

4. Literature Review

a. Respiratory Rate Estimation

Background: Clinical Importance of RR in Neonates

Respiratory Rate (RR) is a critical vital sign for determining the health status of preterm infants in the Neonatal Intensive Care Unit (NICU). The normal respiratory range for neonates is typically between **30-60 breaths per minute (0.5-1.5 Hz)**. Deviations from this range, or the cessation of breathing (apnea), are common in premature infants and require immediate intervention. Currently, the "Gold Standard" for monitoring is contact-based methods, such as Chest Impedance Pneumography (CI) or Electrocardiogram (ECG) leads. While accurate, these adhesive sensors can damage the fragile skin of preterm infants, causing varying degrees of skin trauma, increasing the risk of infection, and causing stress that hinders development. Consequently, there is a significant clinical demand for robust, non-contact monitoring solutions.

Challenges in Non-Contact Monitoring

Developing a video-based system for neonatal RR presents unique challenges compared to adult monitoring:

- Low Signal Amplitude: The chest displacement of a premature infant during breathing is extremely subtle (often sub-millimeter), making it difficult to detect with standard RGB cameras.
- Motion Artifacts: Infants engage in spontaneous, erratic body movements (gross motion) that generate high-amplitude noise, easily masking the delicate breathing signal.
- Environmental Constraints: The NICU environment is complex, with varying lighting conditions, blankets, and medical equipment often occluding the region of interest (ROI).

Different Approaches to Solving the Problem

Research in respiratory estimation has evolved through several distinct methodological phases.

A. Classical Computer Vision and Signal Processing Early approaches relied on deterministic algorithms to extract signals without training data.

- Eulerian Video Magnification (EVM): This technique amplifies subtle temporal variations in video intensity. It has been successfully used to visualize breathing in adults.
- Optical Flow: Methods tracking the movement of specific feature points on the chest over time.
- Pros: These methods are computationally light and require no training data.
- Cons: They are highly sensitive to noise. Classical filters (FFT, Band-pass) struggle to distinguish between respiratory motion and random body movements, leading to high error rates in realistic NICU settings [2].

B. Probabilistic and State-Space Models To handle noise better, some researchers applied Bayesian frameworks or Kalman Filters to track the respiratory waveform.

- Pros: Better handling of uncertainty than simple filtering.
- Cons: These models often assume linear dynamics or Gaussian noise, which does not always hold true for the erratic motion patterns of infants [2].

C. Deep Learning Architectures (Current State-of-the-Art) Recent advancements have shifted towards data-driven Deep Learning (DL) models that learn to extract features directly from video frames [1, 6].

- CNNs and 3D-CNNs: Convolutional Neural Networks (CNNs) are used to detect Regions of Interest (ROI) such as the chest or abdomen. 3D-CNNs extend this by processing a "volume" of video frames to capture motion over time.

- RNNs and LSTMs: Recurrent Neural Networks have been paired with CNNs to model the temporal sequence of breathing. For example, hybrid models combining 3D-CNNs with LSTMs have achieved promising results on thermal video data, with errors as low as 1.6 breaths per minute [4].
- Pros: DL models are significantly more robust to noise and varying lighting conditions than classical methods. They can learn to ignore non-respiratory motion.
- Cons: These models are "data-hungry," requiring large, labeled datasets which are scarce in the medical domain. Additionally, standard CNNs have a limited "receptive field" in time, meaning they may struggle to capture long-term temporal dependencies or very slow breathing patterns. This limitation has paved the way for Transformer-based architectures, which utilize self-attention mechanisms to model global temporal dependencies more effectively than RNNs [3].

b. Body Temperature Estimation

Search Strategy

Literature was collected from IEEE Xplore, PubMed, Scopus, and Google Scholar, prioritizing recent studies (2019–2025). Search terms included neonate/preterm/NICU, temperature monitoring, infrared thermography/thermal video, non-contact monitoring, ROI detection/tracking, multimodal fusion, CPTD, and transformer/attention.

Clinical Motivation and Key Challenges

Temperature is a critical NICU vital sign, and deviations may indicate infection, physiological instability, or environmental stress [8]. Typical neonatal temperature is

approximately **36.5-37.5 °C**, values over 40 °C may indicate fever (or measurement error), while over 45 °C is generally non-physiological and often reflects sensing/estimation artifacts [8]. Contact probes can irritate fragile skin and may be displaced, motivating non-contact approaches [8].

Most camera-based methods estimate skin temperature, whose relation to core temperature depends on perfusion and environmental conditions [8]. In incubators, airflow, humidity, reflections, and geometry can bias infrared measurements, making calibration/compensation important [9]. Motion and occlusions (blankets, caregiver interventions) further complicate stable ROI tracking and continuous estimation [10].

Approach A: Infrared Thermography (Thermal Video)

Infrared thermography (IRT) is the most established non-contact approach because it directly senses emitted IR radiation from the skin [7]. Classical pipelines select and track an ROI (face/forehead/torso) and compute a robust statistic over time, but accuracy degrades under incubator variability and changing geometry [9]. Calibration and environmental compensation improve agreement with references in neonatal incubators [9]. Continuous thermal video systems support trend monitoring [11], and neonatal studies show slow thermoregulatory dynamics that benefit from temporal modeling rather than isolated snapshots [12].

Pros: Direct thermal sensing, continuous trends feasible [7],[11].

Cons: Cost, calibration needs, incubator sensitivity, skin (not core) temperature [7],[9].

Approach B: Learning-Based Robustness and Multimodal Fusion

Deep learning is increasingly used to automate ROI detection/tracking and reject artifact frames, improving robustness in realistic NICU conditions [10]. A practical direction is RGB–thermal fusion, where RGB supports reliable landmark/ROI localization and thermal video provides temperature measurement, increasing stability under posture changes and partial occlusions [13].

Pros: Better ROI stability and robustness [10],[13].

Cons: Data requirements, synchronization/alignment complexity, domain shift [10],[13].

Approach C: Clinically Motivated Thermal Features (CPTD)

Recent work estimates core-peripheral temperature difference (CPTD) from thermal video using automatic ROIs, aiming for a clinically informative measure that can be less sensitive to global drift than absolute temperature [14].

Pros: Clinically meaningful, potentially more robust to offsets [14].

Cons: Requires reliable segmentation and exposed skin [14].

Approach D: Transformer-Based Modeling (Emerging)

Temperature changes can be slow and affected by long-duration disturbances (motion/occlusion), motivating models that integrate longer temporal context. Transformer/attention-based methods are increasingly explored for thermal imagery regression [15] and dynamic thermographic video analysis [16], aligning with the need for stronger spatiotemporal modeling.

Pros: Long-range spatiotemporal modeling [15],[16].

Cons: Higher data/compute demands than classical approaches [15],[16].

Summary and Gap

IRT is the most mature neonatal non-contact technology but remains limited by cost, calibration, and incubator sensitivity [7],[9]. Learning-based ROI automation and RGB–thermal fusion improve robustness for real-world NICU deployment [10],[13]. CPTD offers an additional clinically motivated target [14]. Transformer-based modeling is a promising direction for handling long-range temporal dependencies in thermographic video [15],[16].

5. Project Parameters

a. Assumptions

Environmental and Optical Assumptions:

Sensor Synchronization: It is assumed that the script-based synchronization between the dual RGB monitors and the NIR/IR cameras ensures perfect frame-alignment, which is fundamental for accurate multi-modal data fusion and temporal analysis.

Visibility of ROI: We assume that throughout most of the monitoring period, at least one critical region (face and chest/abdomen) remains visible to the cameras and is not fully obscured by blankets, medical equipment, or staff.

Lighting Stability: It is assumed that lighting conditions in the NICU are stable enough for the CNN model to extract consistent spatial features without sudden gain changes unrelated to physiological factors.

Camera Positioning: The system assumes cameras are mounted at angles that allow for clear anatomical identification (Face & Torso) and remain physically stable to avoid vibrations that could be misinterpreted as respiratory movement.

Physiological Assumptions:

Vital Sign Representation: We assume that the infant's respiratory rate is reflected in mechanical chest movements that can be enhanced by the Motion Magnification algorithm.

Thermal Correlation: It is assumed that there is a high correlation between the temperature measured via IR/NIR and the infant's core body temperature, and that the model can learn this relationship without direct contact.

Metric Dynamics: We assume that respiration and temperature change within timeframes that the Transformer can model, specifically, that respiration is relatively fast (0.5-1.5 Hz) while temperature changes more gradually.

Ground Truth Assumptions:

Clinical Monitor Accuracy: Since the Ground Truth is extracted from clinical monitors, we assume the values displayed on the medical monitor are the "Gold Standard" for model validation.

Synchronization Consistency: Although synchronization is managed by a dedicated script, we assume the latency between the monitor reading and the visual manifestation in the video is constant and negligible for training purposes.

b. Constraints

Real-Time Processing: The system is constrained by the GPU power required to run the Transformer and Magnification models simultaneously. Latency must be kept to a minimum for effective monitoring.

Data Privacy: Due to the NICU environment, there are strict limits on storing identifiable video data. The system must operate on anonymized data or use local edge processing.

Field Conditions: The system is limited by the cameras' line-of-sight. Any prolonged physical obstruction of the infant (for example, during medical intervention) will result in a temporary loss of monitoring.

c. Risks

Extreme Lighting Fluctuations: Sudden transitions between daylight and nighttime lighting in the hospital may affect the accuracy of the NIR to Estimated-IR conversion.

Motion Artifacts: Significant body movements (other than breathing) may cause distortions in the Magnification algorithm, potentially leading to false alarms.

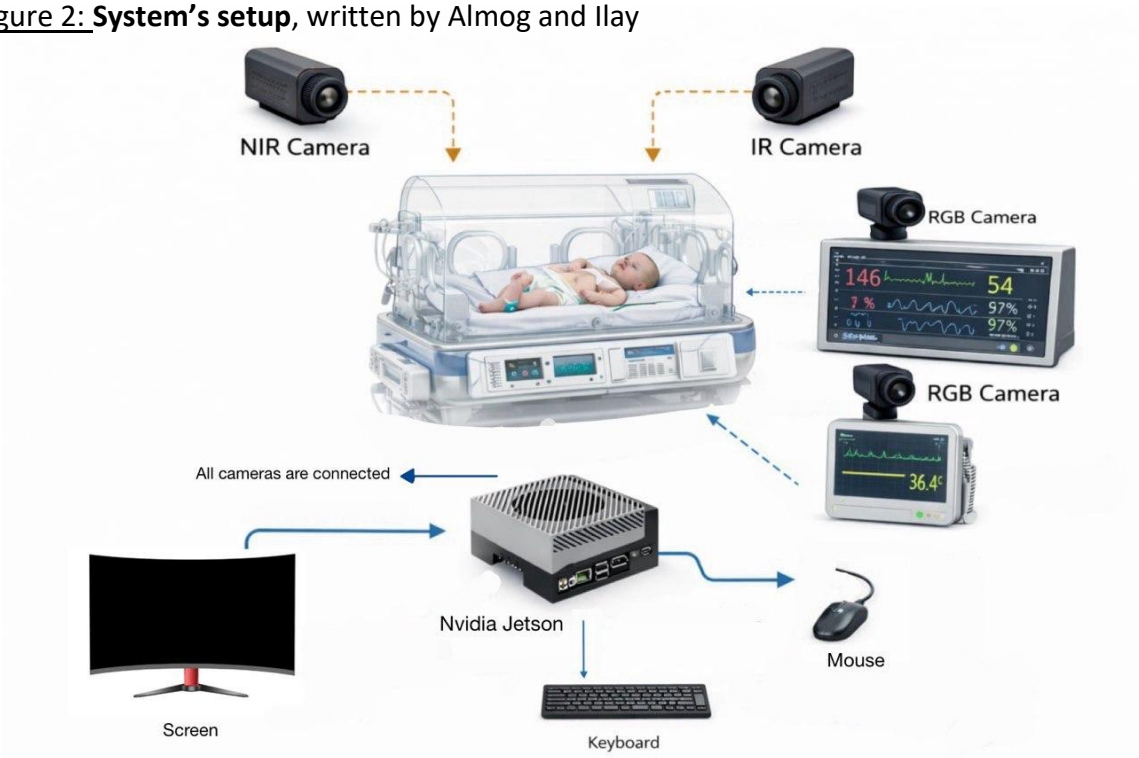
Data Bias: If the training dataset is limited to a specific group of neonates or environmental conditions, the model may struggle to generalize to different cases.

6. Equipment and Budget

- Cameras:
    - E-con systems - IR camera - for the patient
    - E-con systems - CU30 - for the patient
    - E-con systems - ecam-82 - for the Vital Signs monitor
    - E-con systems - cu27 - for the Temperature monitor
- Nvidia Jetson AGX ORIN developer kit
- Mouse and keyboard
- Camera stands
- Essential cables

Equipment	Price	Quantity
IR camera	285 $	1
CU30	152 $	1
ecam-82	110 $	1
cu27	200 $	1
Nvidia Jetson	2500 $	1
Camera stands	10 $	2
Total	**3267 $**	

Figure 2: **System's setup**, written by Almog and Ilay



7. Gantt Chart

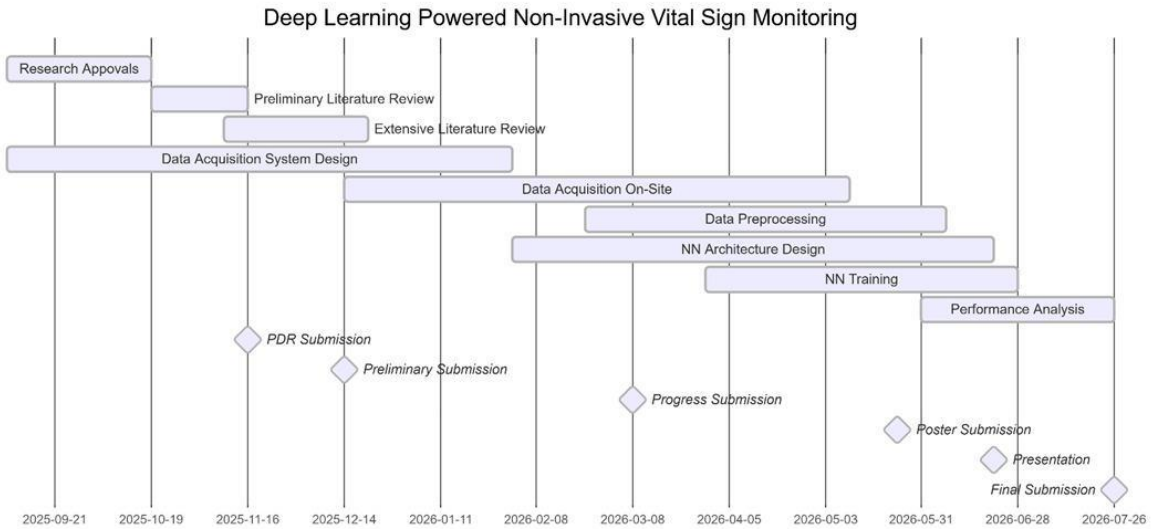


Figure 3: Deep Learning Powered Non-Invasive Vital Sign Monitoring **Gantt Chart**

In a table format -

Task	Weeks
Research Approvals	6
Preliminary Literature Review	4
Extensive Literature Review	6
Data Acquisition System Design	21
Data Acquisition On-Site	21
Data Preprocessing	15
NN Architecture Design	20
NN Training	13
Performance Analysis	8

Milestone	Date
PDR Submission	2025-11-16
Preliminary Submission	2025-12-14
Progress Submission	2026-03-08
Poster Submission	2026-05-24
Presentation	2026-06-21
Final Submission	2026-07-26

Figure 4: **Table format** for Tasks & Milestones

8. References

Respiratory Rate References

[1] S. K. R. Manne, S. Zhu, and M. Wan, "Automatic Infant Respiration Estimation from Video: A Deep Flow-based Algorithm and a Novel Public Benchmark," in *Proceedings of the 26th International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Vancouver, Canada, 2023, pp. 1-10.

[2] M. Bennett, A. Cadik, and R. Bowden, "Video-Based Respiratory Rate Estimation for Infants in Low-Resource Settings," in *IEEE International Conference on Healthcare Informatics (ICHI)*, Houston, USA, 2023, pp. 52-58.

[3] N. Koolen, O. Decroupet, and A. Dereymaeker, "Automated Respiration Detection from Neonatal Video Data," in *International Conference on Pattern Recognition Applications and Methods (ICPRAM)*, Lisbon, Portugal, 2015, pp. 164-169.

[4] Y. Sun, M. Hu, and V. Azorin-Peris, "Motion-compensated non-contact imaging photoplethysmography to monitor cardiorespiratory status during exercise," *Journal of Biomedical Optics*, vol. 16, no. 7, pp. 077010, 2011.

[5] A. Al-Naji, A. G. Perri, and J. Chahl, "Respiratory Rate Estimation from Thermal Video Data Using Spatio-Temporal Deep Learning," *Sensors*, vol. 24, no. 19, pp. 6386, 2024.

[6] W. Verkruysse, L. O. Svaasand, and J. S. Nelson, "Remote plethysmographic imaging of skin variations," *Optics Express*, vol. 16, no. 26, pp. 21434-21445, 2008.

Temperature References

[7] Topalidou, A., Roumeliotaki, T., Lyra, S., Stamatakis, A., & Papageorgiou, P. (2019). Thermal imaging applications in neonatal care: A scoping review. *BMC Pregnancy and Childbirth*, 19(1), 381.

[8] Lei, D., Chai, Y., & Zhang, Y. (2021). Temperature monitoring devices in neonates. *Frontiers in Pediatrics*, 9, 660935.

[9] Hamada, K., Nagasaka, Y., Okada, Y., & Tanaka, S. (2022). Improving the accuracy of infrared thermography in a neonatal incubator. *Annals of Biomedical Engineering*, 50(6), 676–687.

[10] Villarroel, M., Guazzi, A., Jorge, J., Davis, S., Watkinson, P., Green, G., & Tarassenko, L. (2019). Non-contact physiological monitoring of preterm infants in the neonatal intensive care unit. *npj Digital Medicine*, 2, 128.

[11] Lin, J. W., Wang, Y. T., & Chen, C. Y. (2019). A thermal camera based continuous body temperature measurement system. In *Proceedings of the IEEE International Conference on Computer Vision Workshops (ICCVW)*, 302–309.

[12] Morimoto, A., Iwasaki, A., Kawai, R., & Tanaka, S. (2024). Continuous monitoring using thermography can capture heat oscillations maintaining body temperature in neonates. *Scientific Reports*, 14, 1123.

[13] Lyra, S., Stamatakis, A., Papageorgiou, P., & Topalidou, A. (2022). Camera fusion for real-time temperature monitoring of neonates using deep learning. *Medical & Biological Engineering & Computing*, 60(8), 2235–2247.

[14] Zhang, N., Liu, H., Wang, J., & Chen, X. (2025). Automatic and continuous measurement of core–peripheral temperature difference in neonates using thermal imaging. *Biomedical Optics Express*, 16(2), 945–960.

[15] Devi, M. S., Kumar, R., & Singh, P. (2025). Thermal imagery temperature gradient analysis using a vision transformer-based regression network. *IEEE Access*, 13, 45521–45533.

[16] Zhang, Y., Li, Q., & Wang, S. (2025). Video transformer-based analysis of dynamic infrared thermographic sequences. *Infrared Physics & Technology*, 134, 104891.